Andreas Garcia

Backend Engineer

andreas.garcia@themusictree.org · +33 7 70 31 81 05

ABOUT

Engineer with a computer science degree, I work where data, code and clients meet. At Cleversys & Deloitte, I took BI projects from client workshops to Tableau dashboard presentations, and acted as technical point of contact on an ERP migration for a client with €4B in revenue. At INRAE, I helped researchers share their data on an agrivoltaics project, which turned climate change mitigation into a lasting interest.

I founded TheMusicTree, where I build TheMusicTreePipelines, Python and Polars pipelines ingesting MusicBrainz and Wikidata with data-quality checks. They feed GrowTheMusicTree, a D3.js genre map backed by 30.9M songs in PostgreSQL. I run the whole stack myself, from Ansible provisioning to monitoring, and analyse the resulting genre graph in Jupyter notebooks.

SKILLS

Tech: Python, SQL, PostgreSQL, Polars, DuckDB, Jupyter, D3.js, Tableau, Docker, CI/CD
Workflow: Data Pipelines, Data Quality Checks, Data Visualisation, Client Workshops, Specification
Languages: French (native), English (advanced), Spanish (advanced), German (basic)
Hobbies: WWOOFing, natural building (1 year), organizer of a music festival (80+ attendees)

EXPERIENCE

TheMusicTree - Founder & Developer Dec. 2021 – Present · Paris, France

- Building TheMusicTreePipelines, Python/Polars pipelines ingesting MusicBrainz data and the live Wikidata SPARQL endpoint, with data-quality checks and a 90% coverage threshold in CI.
- Building GrowTheMusicTree, an interactive D3.js map of 6,000+ genres whose Django REST Framework and PostgreSQL backend loads 30.9M songs through staged bulk imports from the pipeline outputs.
- Operating the ecosystem myself: Ansible and GitHub Actions provisioning, orchestrated pipeline runs, Prometheus and Grafana monitoring, and offsite backups to Cloudflare R2 with a restore drill.

INRAE - Research Engineer Oct. 2018 – Jan. 2020 · Montpellier, France

- Helped researchers share their data through a common vocabulary and structure on a cross-disciplinary agrivoltaics project, combining agriculture and photovoltaics to mitigate climate change.
- Contributed to OpenSILEX, an open-source information system making research data FAIR for scientists, manipulating complex graph data in RDF4J and improving the underlying ontologies.
- Implemented event-handling REST services in Java with the DTO pattern, wired into the PHP 7 (Yii2) and JavaScript web layer, within a SCRUM team practising peer code reviews on GitHub.

Cleversys → Deloitte - BI/ERP Consultant Apr. 2015 – Oct. 2017 · Paris, France

- Specified and implemented 62 critical accounting interfaces as complex SQL scripts for a large-scale ERP migration project, ensuring relational data integrity throughout the process.
- Served as technical point of contact for a client with annual revenue exceeding €4B, leading specification workshops and training developers on the ERP editor's ETL tools.
- Authored development guides and production deployment procedures for the ERP migration project's ETL tooling, while tracking and managing version releases of ETL processes.

EDUCATION

La Ruche - Programme Itinéraire (Entrepreneurship Program) 2025 · Paris, France

- Three-month entrepreneurship program with nine workshops (marketing, strategy, legal, etc.).
- Conducted market studies and developed a business plan.

INSA Lyon - Engineering degree in Computer Science 2014 · Lyon, France